Master of Computer Applications

Semester – III

Data Analytics for Business

Experiment No: 7

Title: Creating Financial KPI Dashboards

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Problem Statement

You are a Financial Analyst in a manufacturing company. Senior management wants to track revenue, expenses, profit margins, and budget utilization. This experiment creates an interactive Finance KPI dashboard that helps evaluate the financial performance of the organization, using the Budget vs Actual Financial Dataset.

Github Link: https://github.com/Priyanshnolkha/Data-Analytics-For-Business-Projects/tree/main/Experiement%207%20Financial%20KPI%20Dashboard

Methodology

All KPI cards, charts, gauges, tables, and slicers were arranged into a single interactive Finance KPI dashboard in Power BI. The dashboard includes:

- Total Budget KPI Card
- Total Actual KPI Card
- Budget Variance KPI Card
- Budget Variance % KPI Card
- Budget Utilization % KPI Card
- Actual Spending by Payment Method (Donut Chart)
- Budget vs Actual by Department (Bar Chart)
- Budget Utilization % (Gauge)
- Variance % by Region (Column Chart)
- Budget Utilization % by Category (Column Chart)
- Detailed Budget vs Actual Table (Department, Category, Region-wise)
- Interactive Slicers: Date, Department, Category, Region, and Payment Method

The dashboard enables management to analyze revenue and expense trends, monitor budget utilization, and identify departments or regions with significant variances, supporting data-driven financial decision-making.

Dashboard Images

Image 1st:

It displays the overall Finance KPI dashboard without any filters applied (Date, Department, Category, Region, and Payment Method slicers set to "All").

- Total Budget: ₹795.36M and Total Actual: ₹890.03M, indicating actual spending exceeded the allotted budget.
- Budget Variance: -95M with a Budget Variance % of -0.12 (approximately -12%), showing organization-wide overspending.
- Budget Utilization %: 111.90%, confirming that actual expenditure surpassed the budgeted amount.
- Actual spending by payment method is fairly even across Card, UPI, and other channels (~24–26% each).
- The Budget vs Actual by Department bar chart shows Sales and Marketing among the departments with the largest actual spend relative to budget.
- The detail table lists the Marketing–Training–East combination as the worst variance at -44.16%, while the overall total variance stands at -11.89%.

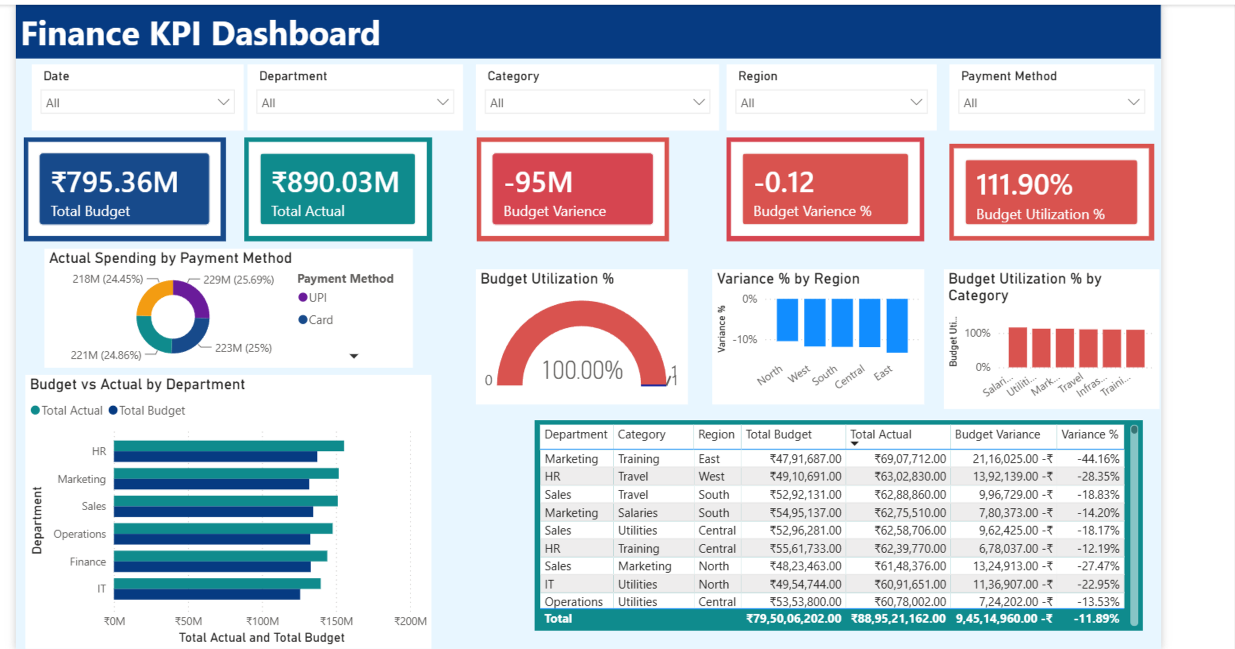

Figure 1: Finance KPI Dashboard – Overall View (No Filters Applied)

Image 2nd:

In the dashboard, after selecting the Operations bar in the Budget vs Actual by Department chart:

- All visuals are cross-filtered to show data for the Operations department only, and the Training category bar is highlighted in the Budget Utilization % by Category chart as Operations' dominant spending category.
- The KPI cards update to Total Budget = ₹5.74M, Total Actual = ₹5.35M, Budget Variance = 381K, Budget Variance % = 0.07 (7%), and Budget Utilization % = 93.35%.
- The detail table now lists only Operations rows — for example, Operations–Training–South shows a Total Budget of ₹57,35,005 against a Total Actual of ₹53,53,635, giving a positive Budget Variance of ₹3,81,370 (Variance % = 6.65%), one of the few rows where actual spend came in under budget.
- This interaction helps analyze budget utilization and spending efficiency specifically within the Operations department.

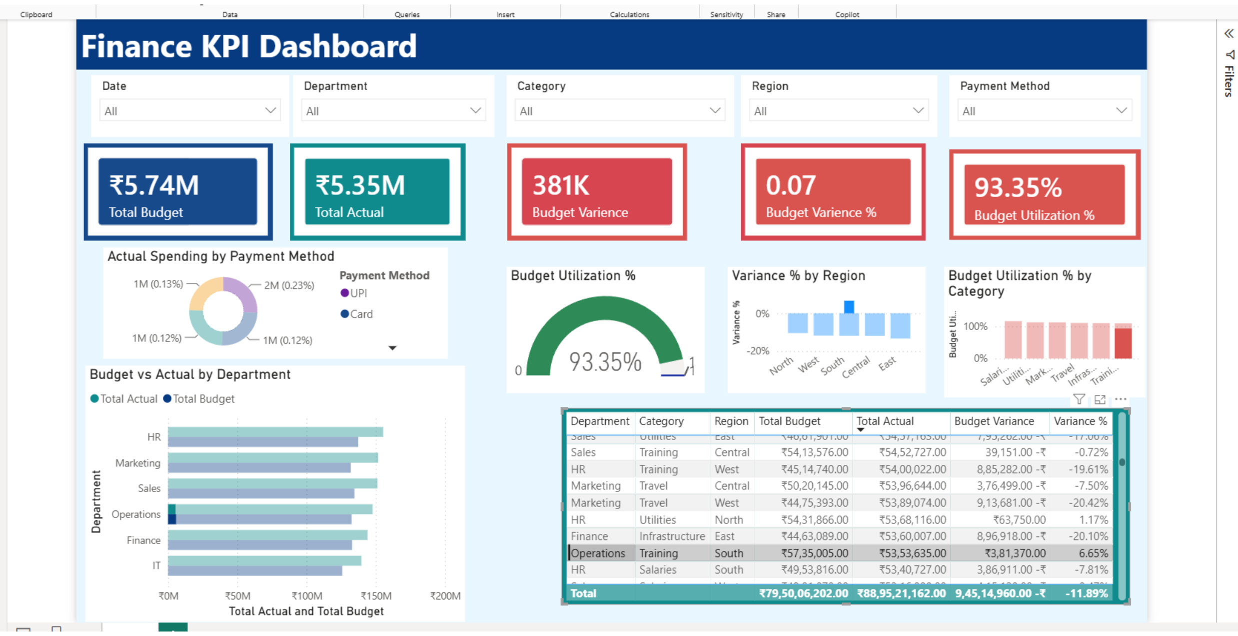

Figure 2: Finance KPI Dashboard – Filtered by Operations Department

Image 3rd:

In the dashboard, after selecting the Finance bar in the Budget vs Actual by Department chart:

- All visuals are cross-filtered to show data for the Finance department only, and the Travel category bar is highlighted in the Budget Utilization % by Category chart.
- The KPI cards update to Total Budget = ₹6.22M, Total Actual = ₹5.14M, Budget Variance = 1M, Budget Variance % = 0.17 (17%), and Budget Utilization % = 82.56% — the lowest utilization among the departments reviewed, meaning Finance spent well below its allotted budget overall.
- The detail table now lists only Finance rows, revealing offsetting variances within the department: Finance–Training–Central overspent (Total Budget ₹41,89,447 vs Total Actual ₹51,73,530, Variance % = -23.49%), while Finance–Travel–North underspent (Total Budget ₹62,22,912 vs Total Actual ₹51,37,405, Variance % = 17.44%).
- This interaction shows that although Finance is under-utilized overall, spending patterns vary significantly by category and region within the department.

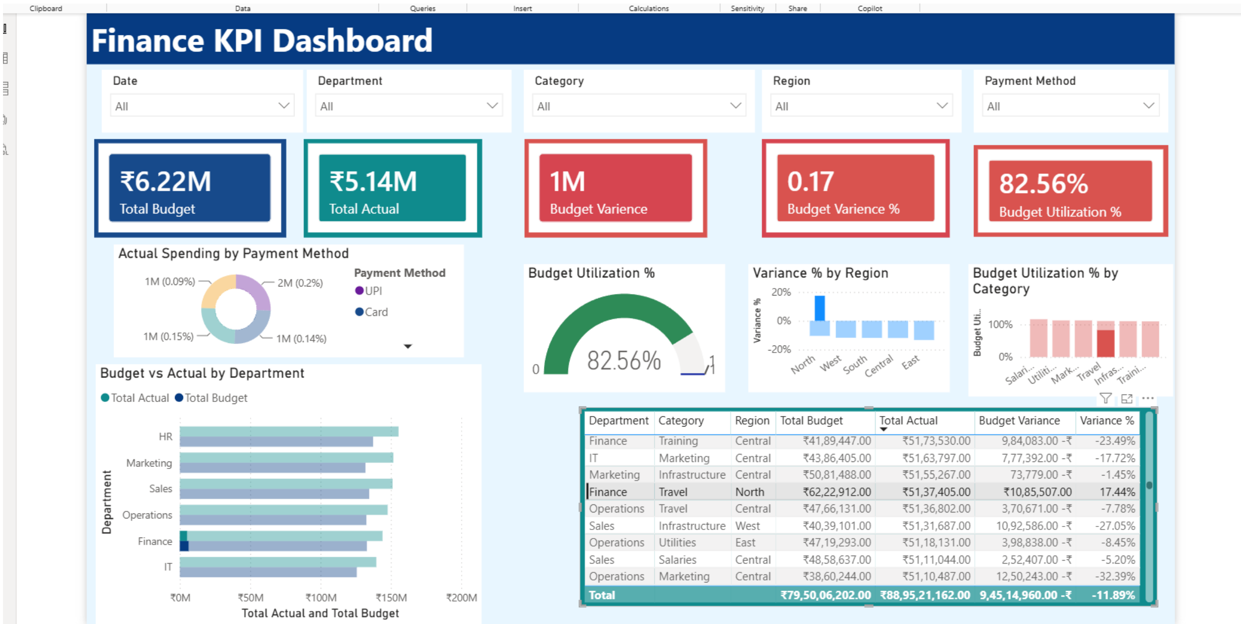

Figure 3: Finance KPI Dashboard – Filtered by Finance Department

DAX Measures Used

The following DAX measures were created on the Budget vs Actual data table to power the KPI cards and visuals in the dashboard:

1. Total Budget

```
Total Budget = SUM('Budget vs Actual'[Budget Amount])
```

Calculates the sum of all budgeted amounts across departments, categories, and regions. Used to display the Total Budget KPI (₹795.36M).

2. Total Actual

```
Total Actual = SUM('Budget vs Actual'[Actual Amount])
```

Calculates the sum of all actual spending recorded. Used to display the Total Actual KPI (₹890.03M).

3. Budget Variance

```
Budget Variance = [Total Budget] - [Total Actual]
```

Calculates the difference between total budget and total actual spend. A negative value indicates overspending. Used to display the Budget Variance KPI (-95M).

4. Budget Variance %

```
Budget Variance % = DIVIDE([Budget Variance], [Total Budget], 0)
```

Expresses the budget variance as a percentage of the total budget, using DIVIDE() to safely handle division by zero. Used to display the Budget Variance % KPI (-0.12).

5. Budget Utilization %

```
Budget Utilization % = DIVIDE([Total Actual], [Total Budget], 0)
```

Calculates the proportion of the budget that was actually utilized/spent. Used to display the Budget Utilization % KPI and gauge (111.90%).

Business Insights

- The organization budgeted ₹795.36M but actually spent ₹890.03M, resulting in a Budget Variance of -₹95M and a Budget Utilization of 111.90% — indicating overall overspending.
- The overall Budget Variance % across all department–category–region combinations is -11.89%, confirming a consistent pattern of actual spend exceeding budget.
- Marketing (Training, East region) shows the highest negative variance at -44.16%, followed by HR (Travel, West region) at -28.35%, flagging these as priority areas for cost control.
- Spending by payment method is fairly balanced, with Card, UPI, and other methods each accounting for roughly a quarter of actual spend, suggesting no single payment channel is driving the overspend.
- Interactive slicers for Date, Department, Category, Region, and Payment Method allow management to drill down into specific combinations to pinpoint the source of budget overruns.
- Drilling into individual departments shows very different pictures: Operations runs close to budget (93.35% utilization, +7% variance) while Finance is notably under-utilized (82.56% utilization, +17% variance) but with offsetting over- and under-spends across its categories — showing that department-level totals alone can mask category-level risk.

Conclusion

The Budget vs Actual Financial dataset was successfully imported into Microsoft Power BI and transformed into an interactive Finance KPI dashboard using KPI cards, bar charts, column charts, a donut chart, a gauge, a detailed table, and slicers. The dashboard effectively tracks revenue and expense performance, budget utilization, and variance across departments, categories, regions, and payment methods. Overall, the interactive dashboard supports data-driven financial planning and cost-control decisions by enabling management to monitor and evaluate budget performance efficiently.